

CHANGLING, China

WMO: 540490

Lat: 44.2784N

Lon: 123.9671E

Elev: 190

StdP: 99.06

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.1	(c) -22.9	(d) -31.0	(e) 0.2	(f) -21.7	(g) -29.1	(h) 0.3	(i) -20.1	(j) 9.2	(k) -11.1	(l) 8.1	(m) -9.1	(n) 1.5	(o) 250	(p) 0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.1	(c) 31.8	(d) 20.8	(e) 30.4	(f) 20.6	(g) 29.2	(h) 20.4	(i) 24.5	(j) 28.3	(k) 23.6	(l) 27.3	(m) 22.8	(n) 26.5	(o) 3.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.4	18.6	27.0	22.4	17.5	26.1	21.6	16.7	25.4	75.7	28.3	72.2	27.4	68.9	26.5	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.7	(c) 6.8	(e) -27.1	(f) 34.1	(g) 3.1	(h) 1.8	(i) -29.4	(j) 35.4	(k) -31.2	(l) 36.4	(m) -33.0	(n) 37.4	(o) -35.3	(p) 38.7		
(4) 9.0	(5) 7.7	(6) 6.8	(7) -27.4	(8) 25.5	(9) 3.1	(10) 1.0	(11) -29.6	(12) 26.2	(13) -31.4	(14) 26.8	(15) -33.1	(16) 27.4	(17) -35.3	(18) 28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-14.5	-9.5	-1.2	8.6	16.2	21.9	24.2	22.4	16.7	8.0	-3.2	-12.3		
	DBStd	14.31	4.74	5.42	5.59	4.99	4.12	3.18	2.31	2.64	3.72	4.95	6.10	4.94		
	HDD10.0	2925	759	546	348	84	3	0	0	0	2	96	397	691		
	HDD18.3	4783	1017	779	604	293	93	6	1	3	72	320	646	949		
	CDD10.0	1659	0	0	1	43	194	357	439	386	204	36	1	0		
	CDD18.3	477	0	0	0	1	26	113	181	131	24	1	0	0		
	CDH23.3	3802	0	0	0	18	259	986	1455	913	166	6	0	0		
	CDH26.7	1070	0	0	0	2	67	341	429	212	20	0	0	0		
(14) Wind	WSAvg	3.0	2.5	2.8	3.6	4.0	3.8	3.0	2.7	2.5	2.7	3.1	3.0	2.6		
(15) Precipitation	PrecAvg	457	1	3	8	18	38	68	158	95	42	18	6	2		
	PrecMax	716	5	13	28	121	102	114	291	216	121	64	21	13		
	PrecMin	277	0	0	0	1	6	19	69	17	5	0	0	0		
	PrecStd	106	1	4	9	23	22	27	62	56	23	16	5	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	7.0	17.5	25.6	30.8	34.7	33.4	31.8	28.8	23.9	14.1	4.1		
		MCWB	-2.5	1.0	8.0	12.4	16.4	19.2	23.0	22.9	19.1	14.7	8.5	2.2		
	2%	DB	-3.1	4.0	12.8	22.4	28.7	32.1	31.7	30.1	26.8	20.3	11.1	0.0		
		MCWB	-5.5	-0.7	5.0	10.8	15.1	18.6	22.5	21.9	18.1	13.0	6.2	-2.2		
	5%	DB	-5.2	1.6	10.1	19.9	26.4	30.4	30.4	28.9	25.2	17.9	8.2	-2.5		
		MCWB	-7.3	-2.2	3.4	9.8	14.5	18.5	22.3	21.7	17.1	11.3	4.5	-4.7		
	10%	DB	-7.3	-0.7	7.6	17.5	24.0	28.6	29.1	27.7	23.7	15.8	5.5	-5.0		
		MCWB	-9.0	-3.8	1.7	8.6	13.2	18.3	22.0	21.1	16.1	10.3	2.0	-6.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.5	1.9	8.6	14.3	18.7	22.5	26.1	25.4	21.9	16.4	9.8	2.1		
		MCDB	-0.2	5.6	15.4	21.3	27.1	28.7	30.7	29.0	26.3	20.4	12.9	3.5		
	2%	WB	-5.5	-0.3	5.8	12.3	17.0	21.4	24.9	24.3	20.3	14.4	7.2	-1.9		
		MCDB	-3.4	3.5	11.7	20.1	24.7	27.4	28.7	27.9	24.4	18.8	9.8	-0.2		
	5%	WB	-7.3	-2.0	3.8	10.7	15.6	20.6	24.0	23.5	19.0	12.9	4.7	-4.7		
		MCDB	-5.3	1.6	9.6	18.4	23.4	26.4	28.0	26.9	22.7	16.6	7.9	-2.6		
	10%	WB	-9.0	-3.9	2.1	9.1	14.5	19.8	23.3	22.7	17.7	11.1	2.2	-6.8		
		MCDB	-7.3	-0.7	7.3	16.4	22.1	25.4	27.1	25.9	21.8	15.0	5.3	-5.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.4	10.4	11.4	11.0	9.6	8.1	8.6	10.9	10.3	9.0	8.9		
		MCDBR	10.9	11.8	13.3	14.4	13.8	12.6	9.9	10.0	12.1	12.4	10.9	10.2		
		MCWBR	9.1	8.7	7.9	7.3	5.7	4.4	3.6	4.1	5.3	6.7	7.6	8.5		
		MCDBR	10.7	11.6	12.4	12.3	11.5	9.1	7.5	7.6	9.4	10.6	10.5	10.1		
	5% WB	MCWBR	9.0	8.7	7.7	7.1	5.6	3.8	3.7	4.0	5.3	6.7	8.0	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.339	0.410	0.511	0.587	0.651	0.605	0.537	0.471	0.453	0.369	0.323		
	taud		2.138	2.095	1.960	1.772	1.666	1.625	1.765	1.910	1.985	1.920	2.056	2.095		
	Ebn at Noon		782	827	808	754	710	667	693	727	740	678	690	713		
	Edh at Noon		96	121	159	210	241	252	216	181	155	144	105	91		
(44) All-Sky Solar Radiation	RadAvg		2.21	3.23	4.38	5.32	5.89	5.67	5.38	5.06	4.50	3.22	2.16	1.81		
	RadStd		0.20	0.23	0.22	0.40	0.40	0.47	0.46	0.34	0.31	0.20	0.18	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	-1.07	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page